

TOTAL MARKS: 40

TIME: 60min

Questions 1 to 12 carries 1 mark each.

1. The first general purpose electronic digital computer in the world was
A) UNIVAC B) EDVAC C) ENIAC D) All of the above
2. C is ?
A) A machine language B) A third generation high level language
C) An assembly language D) All of the above
3. _____ holds the contents of the accessed memory word.
4. Which of the following memories needs refreshing?
A) SRAM B) DRAM C) ROM D) All of above
5. The ALU of a computer normally contains a number of high speed storage element called
A) Semiconductor memory B) Registers C) Hard disks D) Magnetic disk
6. The original ASCII code used _____ bits of each byte, reserving that last bit for error checking
A) 5 B) 6 C) 7 D) 8
7. Which statement is valid about interpreter?
A) It translates one instruction at a time B) Object code is saved for future use
C) Repeated interpretation is not necessary D) All of above
8. A computer program that converts an entire program into machine language at one time is called a/an
A) Interpreter B) CPU C) Compiler D) Simulator
9. _____ provides a consistent way of encoding multilingual plain text.
10. To perform product of maxterms the Boolean function must be brought into product of
A) and terms B) or terms C) not terms D) nand terms
11. Convert 1101010_2 to $?_8$.
12. Convert ABC_{16} to $?_2$.
13. Using 2^s complement calculate $100011 \div 000101$ [4]
14. Using truth table prove that $AB+BC+CA' = AB + CA'$ [6]
15. Give the difference between CISC and RISC processors. How is multithreading applied in single core and multi-core processors. [5+3]
16. What is a Flash ROM? Give two examples each of ROM and Flash ROMs. [2+2]
17. What is principle of locality of reference? Explain cache hit and cache miss. [2+4=6]

OR

Why is memory interleaving used? For an n-way interleaved memory, where does the i^{th} memory location reside? What is the significance of the width of the memory address bus?

[2*3=6]